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Drilling Fluids Digitalization: Enabling Real-Time Performance, Automation, and Flexi-Billing Through AI-Ready Data Infrastructure

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Abstract

As the oil and gas industry accelerates its transition toward digitalization, the effective integration of drilling fluids data into centralized, intelligent systems has become increasingly critical. This paper presents the development and successful implementation of a pioneering digital drilling fluids interface, which automates data capture, enhances reporting integrity, and enables real-time analytics—all embedded within the existing drilling database.

The primary objective was to establish the first foundational layer of a fully digital and structured drilling fluids database, designed to support future AI-driven automation, Big Data analytics, and remote operational decision-making. The project successfully delivered both key milestones automated data insertion and extraction—and validated three years of high-integrity fluid records for operational use.

Keywords: Drilling Fluids Digitalization, Enabling data automation, live performance, consumption tracking, fluids design optimisation and instant flexi-billing

Introduction

As mentioned, Solution represents fully digital infrastructure to automate processing, validation and migration the data from external file structures (Excel) into structured Company's master data storage. To connect different stakeholders of drilling fluid reporting activities solution, combine several informational systems into automated data transfer workflow. There are such systems like: Microsoft outlook, Microsoft Excel, Microsoft windows file explorer, data quality integrational layer, master data storage, drilling reporting system and windows active directory. Below is a graphical representation of solution architecture.

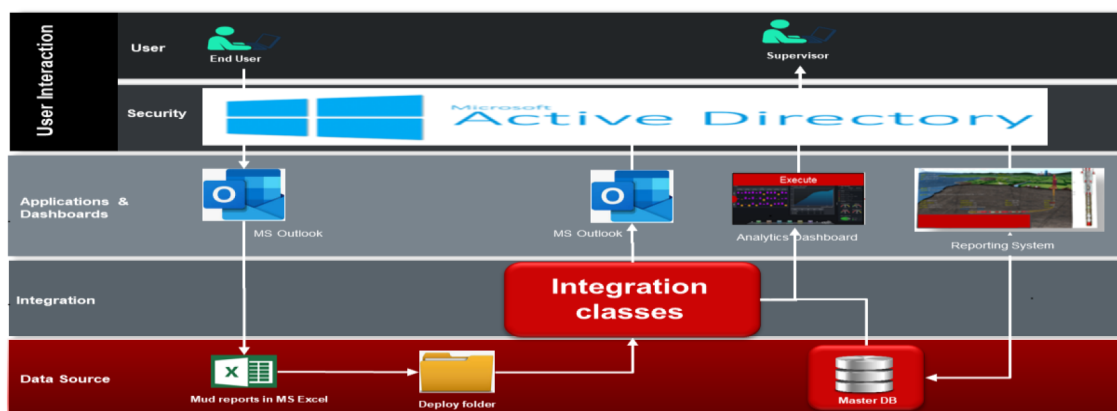


Figure 1—Solution Architecture

The main functional requirements of the Solution have been consolidated into the following specifications sections:

1. Data transfer requirements
2. Excel template data requirements and matches master data Storage structure
3. Data quality requirements
4. Data Entry forms requirements
5. Data quality reports and Analytics dashboards
6. Alerts and notifications
7. System health and integrity monitoring.

Methods, Procedures, and Process

A cross-functional team comprising fluids engineering, IT digital, and third-party software providers co-developed a fully automated interface. The system extracts input from customized Excel-based daily reports, applies a robust quality check, generates instant feedback (accepted/rejected with reasons), and stores verified data into a structured database featured by the current system in place. The workflow was designed in alignment with the Operator's SOPs, ensuring compliance with drilling operations governance. Additionally, a Spotfire-based dashboard was built to enable multi-dimensional analytics, tracking both operational performance

Solution Development

Development of the Solution consists of several steps:

1. Solution design – series of collaborative sessions where different subject matter experts were discussing the conceptual steps and data transfer approaches
2. Automated procedures development:
 - a. Extracting the excel reports from outlook SMPT server
 - b. Transfer excels files to deploy Folder
 - c. Data processing and extraction from excel template
 - d. Applying data quality rules on the processed data (100+ data quality rules)
 - e. Generating automated messages in case of data quality issues
 - f. Matching data between excel template and master data storage
 - g. Transferring data to master data storage
 - h. Adjusting data entry forms in existing reporting system

3. User Interface Development

- a. Generating automated pop-up messages in case of data quality issues
- b. Development of SpotFire dashboards
- c. Creating output reports according to company's standards
4. User acceptance test: Set of use cases to be performed in the solution to validate compliance with designed concept
5. Solution Implementation: deployment of solution architecture in the company's production environment.

At the end it has been decided to consider 2 types of data migration workflows: Full automated and Semi-automated methods.

Full Automation

This is the primary workflow for the solution, and it assumes an automatic approach of data extraction, processing and transferring them into company's master data storage. It will consist of several steps as shown below and illustrated in [figure 2](#):

6. Business User at the company rig sends the daily mud report in *.xlsx format (excel template) to nominated functional mailbox.
7. Once Excel template comes to functional mailbox this will trigger the workflow of data extraction:
 - a. Save information about sender's email
 - b. The file name can be anything because it will be renamed after the process runs.
 - c. File extension is .xlsx
 - d. Save excel file name and location
 - e. Place Excel template to deploy folder
 - f. Extract data from Excel template
 - g. Check if daily operations report object exists in drilling reporting system based on the following information from header section of excel template: report number, well name, rig name and date.
 - h. The report date must match the system date.
 - i. Places excel template into waiting folder daily operations report if object doesn't exist in drilling reporting system and send notification email to sender to create it.
8. Apply Data Quality
 - a. If Excel template has passed through data quality rules, then initiate data transfer process to company's master data storage
 - b. If excel template has NOT passed through data quality rules, then it would be transferred to rejected folder and notification email will be send to sender's email back with list of revealed data quality issues
 - c. Transfer data company's master data storage
 - d. Check that data has been stored in company's master data storage
 - e. Transfer processed excel template to the constant share folder to be accessed by drilling fluid team members

After finishing the data transfer process, the Solution will send an email to the functional email box so that the workflow has been successfully completed and links to the Spotfire dashboards will be provided.

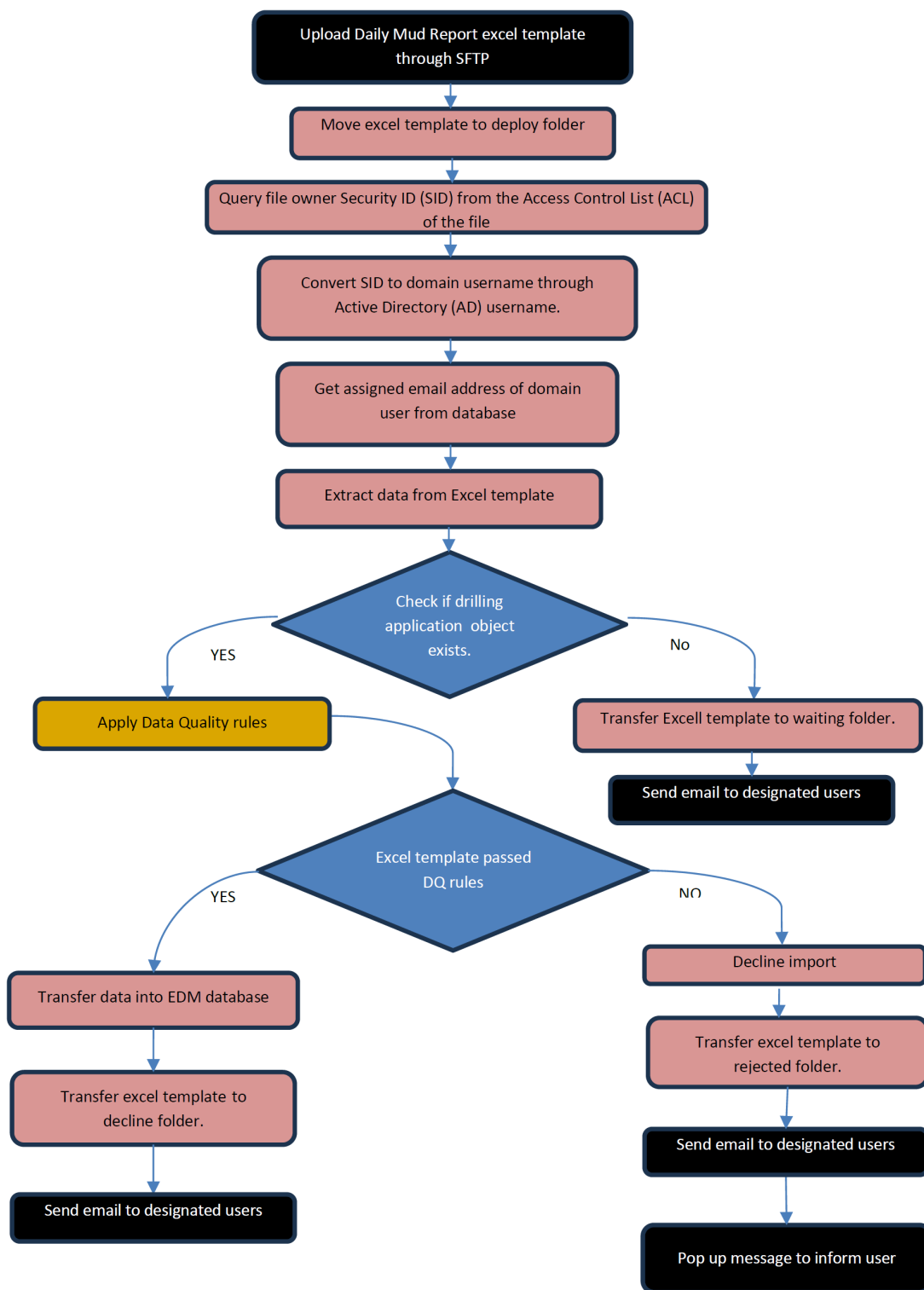


Figure 2—Fluid report full automation steps

Semi-Automated

This workflow is assumed to support fully automated workflow in case of rigs remote operation (outside network) or if some excel templates of mud daily reports have not been automatically transferred to company's master data storage due to rejection action or any other related circumstances which leads to absence of drilling fluid related data in the drilling reporting system.

The workflow consists of several steps as shown below and illustrated in [figure 3](#):

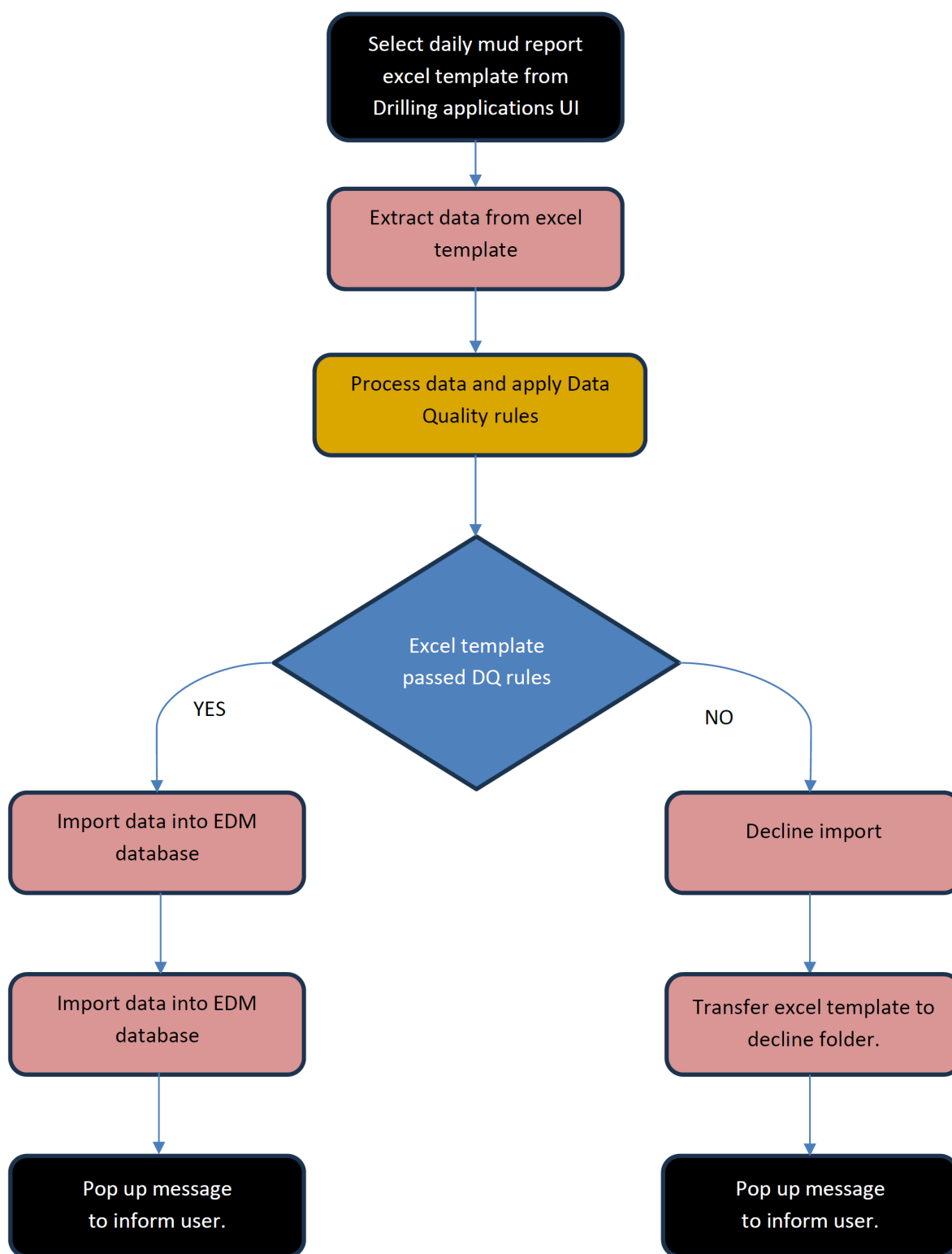


Figure 3—Fluid report semi automation steps

9. Business users from the company's rig or drilling fluid team member locates, and clicks import button in the drilling reporting system

10. Business users from the company's rig or drilling fluid team members locate and select Excel template of daily mud reports
11. Process excels template data and applies data quality rules
 - a. If excel template has passed through data quality rules, then initiate data transfer process to master data storage
 - b. If excel template has NOT passed through data quality rules, then the import procedure will be declined and the pop-up message with revealed data quality issues will appear
12. Fix data quality issues and import data to master data storage again

During data transferring workflows and procedures there is a possibility of getting a similar copy of Excel template for the same well, rig, report number, and date that has been processed earlier. In such evidence the Solution will automatically reject these files. Data overwriting is permitted only for manual import. In the case of manual import the user will receive a pop-up message that data has been already imported, and it should be additionally confirmed to overwrite the existing data.

Execution Challenges

Data Quality check challenges

The key challenge during designing stage is to ensure:

- Data quality (How to ensure data inserted is accurate and reliable, with negligible human errors): taking into consideration that the fluids data were submitted on daily basis by different users, rigs & with different drilling activity.
- Data accuracy: Is how to ensure that the fluid report of rig A does not insert into rig B well. as shown in [figure 4&5](#).

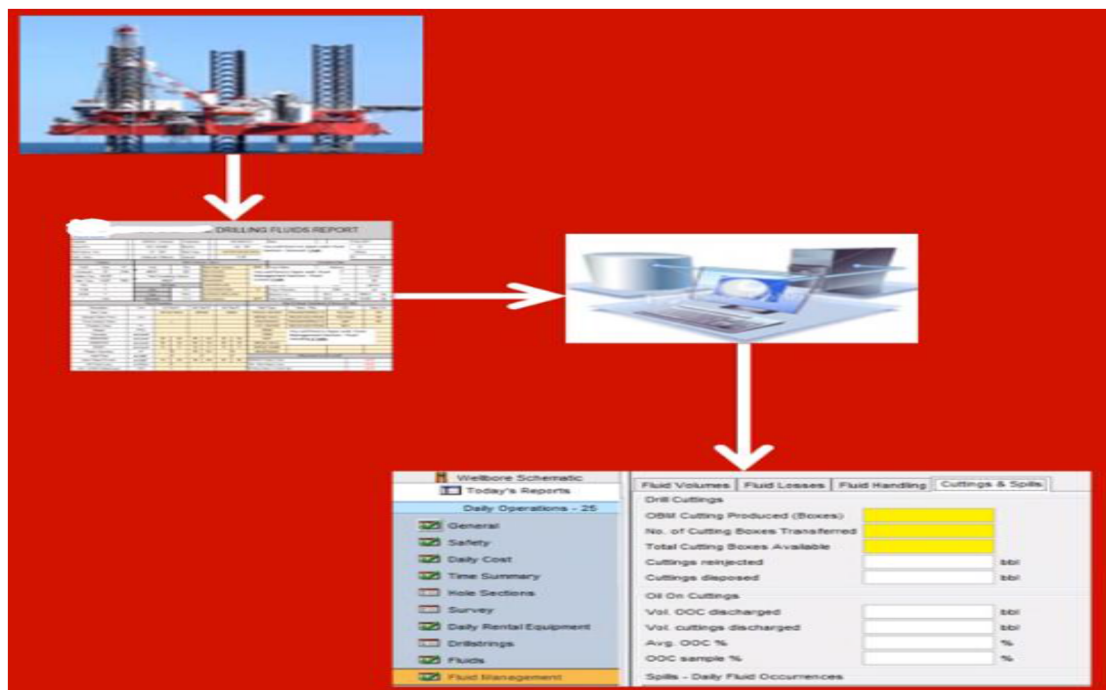


Figure 4—shows data workflow from rigs to our drilling novel database.

workflow for data transferring

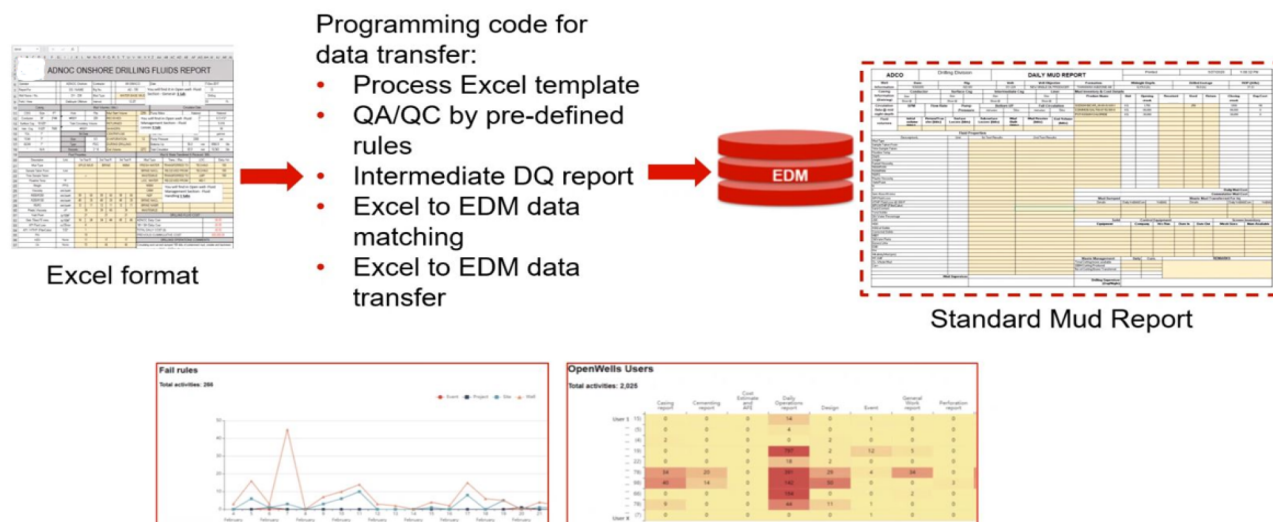


Figure 5—data quality workflow diagram

Therefore, the solution to tackle above was to create a unified excel template to ensure always we have the upper hand and control over users' data entry on the daily fluid reports.

Therefore, the team embedded more than 400 rules on the IT digital interface tool and more than 600 rules on the unified excel report template itself, specifically for all the mapped cells data selected to be imported to our master drilling application.

QA/QC rules

Excel template rules. Defined a set of rules and controls over excel sheet as shown in figure 4 excel data QA/QC, this successfully enabled us to:

1. Ensure all the rig fleets reporting data are 100% unified - for example date format, well number (fixed format), rig name.
2. Daily drilling fluids technical data inserted on daily report by different engineers or users at the rig site.
 - a. Defined a minimum set of cells required from all rigs (drilling, workover, rig-less), if not met the fluid report will be immediately rejected, therefore this rule standardizes a minimum set of data on our database
 - b. Data related to fluid properties, volumes & inventory, utilized excel to set a certain rule for each cell as example defines a capping numbers of users can exceed or set a specific format for the excel cell (users cannot over-right).
 - c. Set and define specific mandatory cells that each single report must have in-order to accept the report, especially for the data the team needed the most based on the operation to capture accurate data and tackle the problem ahead. Example drilling losses: we add a mandatory cell that each user must report, by this we manage to have a digital database on losses and hence build losses map accordingly utilizing varies mapping applications.

Digital interface tool rules. This is a key barrier the team utilizes to customize and screen out inconsistent and failed data to avoid human errors and mistakes. Example:

1. Define specific cells and cross checked them and mapped them with specific cells on our drilling reporting application, if those cells did not match the report will be immediately rejected (these

specific 5 cells are the main barrier to ensure each report inserted into the corrected well number or event level on our drilling database.

2. Define several rules related only to fluids daily inventory table (Inventory is one of the main tables used by our service partners for invoicing at the end of or during the well)
3. Classified invoice to 5 main categories which enable us to customize our drilling database to capture the cost tracking for us based on the category and vendor name. Pricing mechanism and calculation logic were different between all fluids related service. Example the mechanism could be based on consumption, and this require having stock and consume as operation go, where you can receive equipment and charge it based on number of days. Hence all these different calculation mechanisms were taken on consideration when redesigning our novel drilling application to fit our operation needs and contracts obligation.

Cost Plugin design. In-order to capture all related drilling fluids services costs on our novel drilling application, we created a plugin for the mud inventory section which alters the layout and cost calculation logic on this screen based on contract requirements. Figure 6 below shows the inventory section on drilling application divided into three parts

The screenshot displays the 'Mud Inventory' application interface, divided into three main sections:

- Daily Mud Cost Summary:** Located at the top, it shows various cost categories and their values:

Daily mud	9,442 \$	Daily Mud Equipment Costs	247 \$	Daily Mud Fluid Hauling Costs	0 \$
Daily Mud Chemicals Costs	0 \$	Daily Mud Crew Cost	0 \$	Daily Mud Boxes Cost	9,195 \$
- Mud Products:** A table listing different mud products with columns for Category, Product, Units, Price per unit (\$), Vendor, Start, Used, Received, Returned, Adjusted, End, and Daily Cost (\$).

Category	Product	Units	Price per unit (\$)	Vendor	Start	Used	Received	Returned	Adjusted	End	Daily Cost (\$)
1 Cutting Boxes	Box	0.00	0.00	10.00	5.00	0.00	5.00				
2 Equipments	EA	0.00	1.00	0.00	0.00	1.00					
3 Haulage Fluids	BBL	0.00	0.00	0.00	0.00	0.00					
4 Mud Chemicals	55 Gal	0.00	0.00	0.00	0.00	0.00					
5 Personnel	BBLs	0.00	0.00	0.00	0.00	0.00					
- Transactions of the selected product:** A table showing transactions for a selected product, with columns for Used, Day cost (\$), Location A, Trans. / Rec., Location B, Received, Returned, Adjusted, and Transfer Price per Unit (\$).

Used	Day cost (\$)	Location A	Trans. / Rec.	Location B	Received	Returned	Adjusted	Transfer Price per Unit (\$)
1		Al Dhabbia (Onshore)	RECEIVED FROM	TDU	10.00			
2		Al Dhabbia (Onshore)	TRANSFERRED TO	TDU		5.00		

Figure 6—Inventory main sections

- Daily mud cost summary: The most upper part of the screen contains the cost summary by category.
- Master-detail spreadsheet. On the middle part of the screen the users will enter the mud products while on the bottom the transactions must be entered for each mud product.
- The transaction section, where data inserted based on the product category is classified on the background and then calculated based on pricing logic

Mud product uniqueness is determined by the five parameters below

- Category (Chemicals, Equipment, Personal, Haulage Fluid, Cutting Boxes)
- Product
- Units
- Price per unit
- Vendor

Until all the above parameters are fully matched with provided picklist, then the transactions can be inserted for the product in the below part of the screen. If any of the above parameters are different then new mud product must be added. As well as that we redesigned new cost logic in addition to original logic based on selected category.

Mud Products

If users select mud chemicals as category, then both the cost and end stock calculation logic is the same as in the original drilling novel application. Based on the type of selection made in the Category section, the products, Transfer/Receive and Location column will be entered with their respective values. A plugin will be configured to display the daily cost for each category. Logical workflows to calculate the pricing and inventory are followed for each Category, defined as below.

Equipment and Personnel

Cost calculation: Daily cost is the sum of the costs of all the transaction of the given product where the "transfer price per unit" is multiplied by "Used" quantity.

Category	Product	Units	Price per unit (\$)	Vendor	Start	Used	Received	Returned	Adjusted	End	Daily Cost (\$)
Equipments	Air Pump	EA	1.00		0.00	1.00	0.00	0.00	0.00	1.00	1.00

Used	Day cost (\$)	Location A	Trans. / Rec.	Location B	Received	Returned	Adjusted	Transfer Price per Unit (\$)
1.00	1.00							

Figure 7—Inventory tab (Equipment category – cost logic)

In figure 7 above a new inventory logic is implemented whenever the category is equipment or personal and it is used by (Y days) the end stock always increases by the same amount, as example if start stock was X and used (Y days), the end value cumulative carried over is X+Y

Cutting Boxes and Haulage Fluids

Cost calculation: Daily cost is the sum of the costs of all the transaction of the given product where the "transfer price per unit" is multiplied by the received or returned quantity.

Category	Product	Units	Price per unit (\$)	Vendor	Start	Used	Received	Returned	Adjusted	End	Daily Cost (\$)
Cutting Boxes	Cutting Skips	Box	0.000		0.00	0.00	10.00	5.00	0.00	5.00	

Used	Day cost (\$)	Location A	Trans. / Rec.	Location B	Received	Returned	Adjusted	Transfer Price per Unit (\$)
			RECEIVED FROM	TDU	10.00			
			TRANSFERRED TO	TDU		5.00		

Figure 8—Inventory tab (Cutting Boxes category – cost logic)

In figure 8 above as an example: if receiving 10 and transferred 5 the results end of stock will be 5. The daily cost logic will be, if received X; received cost: X*\$ and returned Y; returned costs: Y*\$, and the total daily cost of this product will be (X*\$+Y*\$= Z\$)

Haulage Fluids

The haulage fluids products are designed based on operator requirements and incorporated in the products picklist. So, this enables us to capture the data and fluids movement between location and able to apply the cost associated with it.

Field Execution Results

Digital data Input & Output benefits and values

The digitalization of the drilling fluids workflow demonstrated measurable and strategic benefits across operational, financial, and performance domains:

- +100% reduction in manual data entry, minimizing errors and repetitive tasks
- Eliminate human interference, hence reduce errors probabilities
- Zero cost variance in drilling fluids consumption by aligning product logic with contractual billing schemes
- Seamless integration into legacy drilling databases, enhancing collaboration across stakeholders
- Successfully integrated 100% of our drilling operation onboard utilizing the legacy tool
- Standardized reporting templates improved consistency and reduced onboarding time
- Enable operator to drive fare competition between fluids business partners in-order to improve performance
- Instant and flexible invoicing capability, driven by real-time service data

At the end of the job, our service partners receive their end of well invoice from rig drilling supervisors extracted directly from our drilling application as on [figure 8](#) below and the same used on their billing process. This enabled us to add an extra layer of data accuracy on data-entry level filled by fluids service partners engineers on rig sites, in addition to the rules embedded on initial design on the excel template and interface digital tool. This enabled us to ensure full data accuracy in terms of cost as well as achieving 100% price confidentiality, because as all services related cost were removed from excel template and only kept internally to our drilling novel application.

The output is grouped by product category and vendors for the chemicals and personnel related data. These reports were developed using crystal reports with a supporting java class to generate an input form. This input form enables users to select three input parameters for the report.

Three reports parameters

- Date interval: Specifies the time range for the report. The default start and end values for this parameter are always set to the start date of the event and to the last available report date.
- Inventory categories: Allows selection of which categories to include in the report.
- Vendors: Filters the report to include only the selected vendors.
 - Enabled live performance monitoring, consumption tracking, and KPI-based decision-making

Figure 9—Drilling application customized flexi end of well invoice

Below [figure 10](#), represent the live dashboard extracted from the fluids data added to our drilling database which we stream live daily. Also, we have the privilege to filter and customize the data search on the dashboard to any level either for ongoing or previously completed operation.

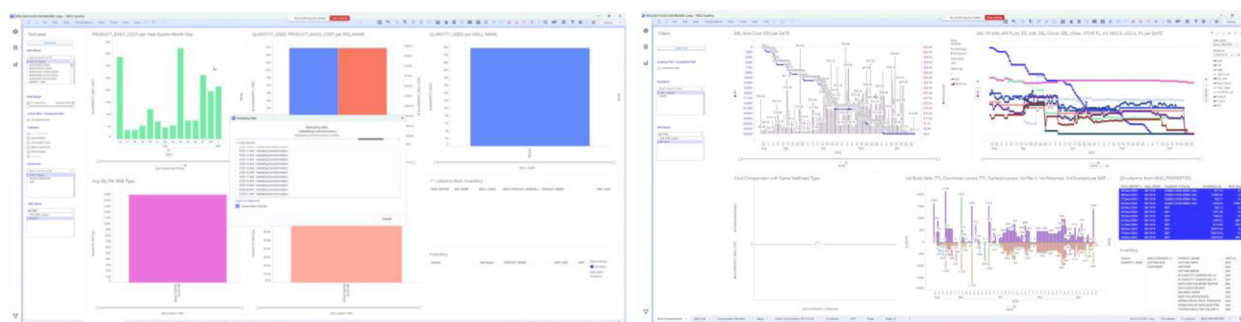


Figure 10—Digital fluids data output dashboard

The above dashboard has several tabs where you can view and customize based on your demands, for example daily drilling fluids operation, ongoing service per vendor, daily cost, stock on fields, daily consumptions (filtered by the highest consumed on the exact search date) and losses daily tracking. These losses data were extracted and integrated with other data and created a field losses map dashboard as shown on [figure 11](#) below, which give operator the upper hand on future design to have necessary mitigations on designing face for upcoming operations and have solids plans to cure it if losses were identified on the same area.

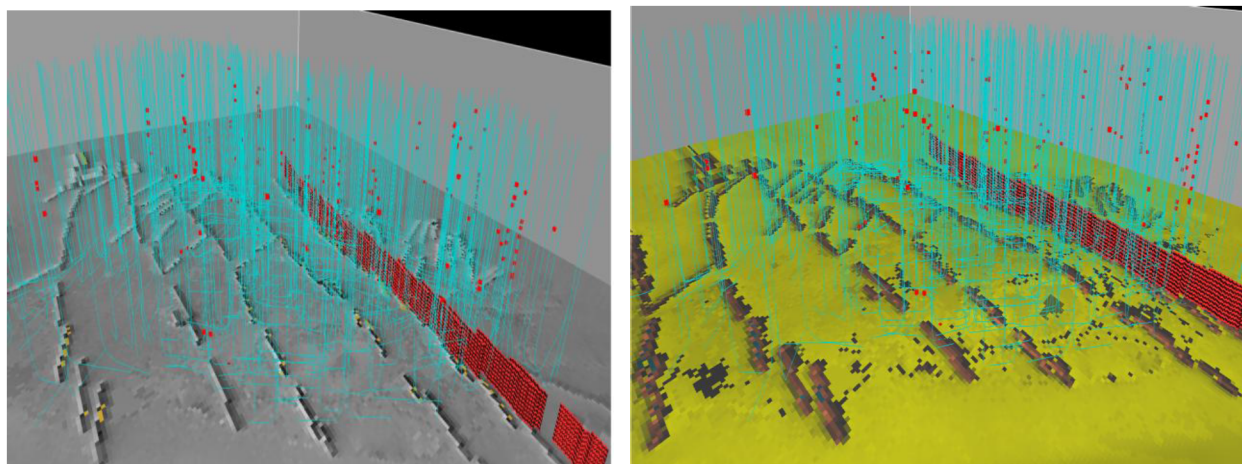


Figure 11—Digital fluids losses map dashboard

- The automated tool processed since launched more than 200 thousand drilling fluids reports with 75% successfully accepted and imported into database as shown on below figure 12.

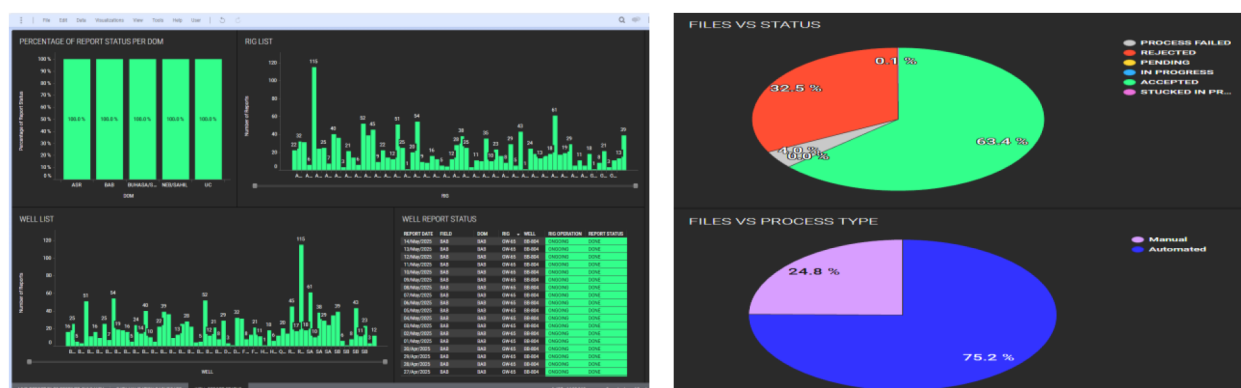


Figure 12—interface tool monitoring dashboard

Conclusions

The solution is the first of its kind in the global oil and gas sector to implement a fully automated drilling fluids data pipeline directly integrated into an enterprise-level drilling database. The approach represents a substantial leap toward the digital oilfield vision, positioning the operation at the forefront of data-driven excellence and future-ready AI integration. This project not only delivered operational efficiency but also laid the groundwork for AI-driven predictive fluids management and remote monitoring capabilities—ushering in a new era of intelligent well operations.

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